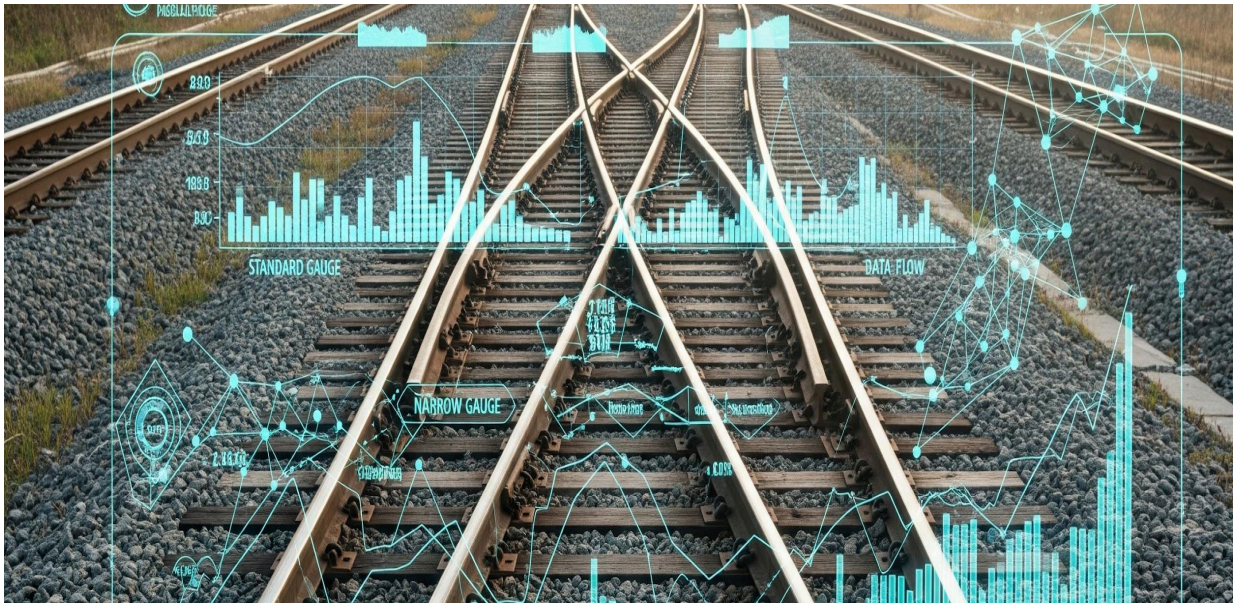




Data Analytics Project: Railway Gauge Trends

Dataset Used: railwaygauge_csv

This document outlines a structured, end-to-end data analysis project based on railway gauge data. The project is divided into sections: **Data Analytics Project: Railway Gauge Trends**

This document outlines a structured, end-to-end data analysis project based on railway gauge data.

Dataset: Railway Gauge Data (Assumed to be in a CSV file)

Project Goal: To analyze historical trends and composition of different railway gauges (Broad, Metre, Narrow) over the years.

Modules Used

The **primary libraries/modules** required for this project are:

- **Pandas:** For all data manipulation, filtering, aggregation, and feature engineering tasks.
- **NumPy:** Specifically for numerical calculations like yearly growth (`np.diff`).
- **Visualization Library (e.g., Matplotlib):** For generating all required plots (Line Graphs, Bar Charts, Pie Charts, Stacked Bar Charts).

Project Coverage

Category	Skills Covered
Data Preprocessing	Data Loading, Missing Value Handling, Type Conversion
Pandas Operations	Filtering, Selection, Aggregation, Feature Engineering
NumPy Calculations	Yearly growth calculation (np.diff)
Visualization	Line Graph, Bar Chart, Pie Chart, Stacked Bar Chart

Scenario 1: Basic Data Loading & Cleaning (Easy)

Description: This scenario focuses on preparing a raw Indian Railways dataset for analysis. You will practice importing data from a CSV file and inspecting its initial structure to identify inconsistencies. The goal is to handle missing information and ensure that numerical columns are correctly formatted for calculations. This foundational step ensures your data is reliable before you perform any deeper insights.

Task	Description
1.	Load the dataset into a Pandas DataFrame.
2.	Display the first 5 rows and column names to understand structure.
3.	Check for missing values and replace them with 0 (assuming missing values represent 0 km of track).
4.	Convert all relevant gauge columns (Broad, Metre, Narrow, Total) to numeric types (e.g., float64 or int64).

Conclusion:

This scenario shows how to turn raw data into a clear visual story using Python. By cleaning the data and creating a line graph, we can easily see how the railway network changed over time. These basic steps are essential for making accurate decisions based on real information. You now have a solid foundation for more advanced data analysis.

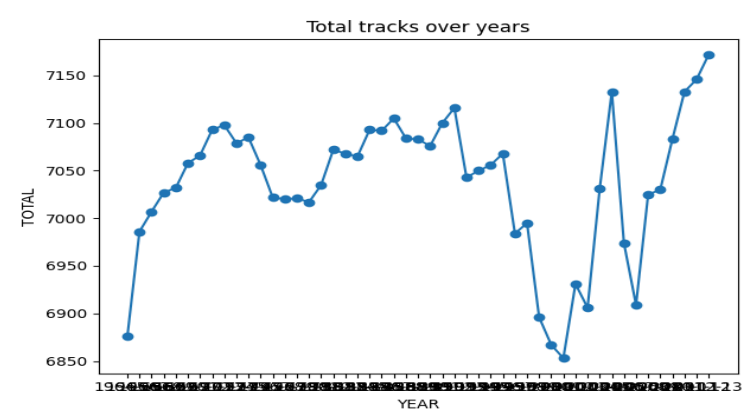
Scenario 2: Simple Visualization (● Easy)

Description:

This part of the project turns numbers into a picture. We pick out the Year and Total columns to see how the railway grew. By drawing a line between these points, we can see if the tracks increased or decreased over time. It is a quick way to understand a long history of data.

Task	Description
1.	Extract the Year and Total columns.
2.	Plot a line graph showing Total tracks over the Years .
3.	Add a descriptive title (e.g., "Growth of Total Railway Network Over Time") and X/Y axis labels.
4.	Identify whether the overall trend is increasing or decreasing and note the finding.

Graph :



Graph Description :

The image shows a line graph titled "Total tracks over years." It tracks the total length of railway lines across many decades. You can see the line rising and falling, which shows that the total track distance didn't just grow—it changed frequently. The dots on the line mark the specific data points for each year.

Conclusion:

This code helps us see the data as a picture instead of just a list of numbers. By making a line graph, we can easily track the growth of the railway over many years. The graph shows that even though there were some drops, the total amount of track is mostly going up. Adding a title and labels makes the information clear and simple for everyone to follow.

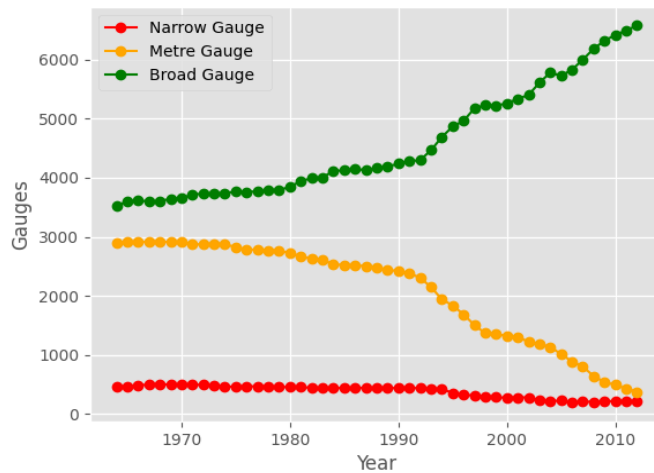
Scenario 3: Filtering + Bar Chart (🟡 Medium)

Description:

This step helps us look at the three kinds of train tracks used after the year 2000. We want to see which type is used the most in modern times. By putting them in a chart together, we can see how the railway is changing. This makes it easy to spot which tracks are growing and which ones are being removed.

Task	Description
1.	Filter the dataset to include only years <i>after</i> 2000.
2.	Select the Broad Gauge, Metre Gauge, and Narrow Gauge columns for the filtered data.
3.	Plot a grouped bar chart comparing the three gauges for the selected years.
4.	Add a legend and ensure proper labels for axes.
5.	Identify which gauge dominates in recent years (post-2000).

Graph :



Graph Description:

The graph uses three colored lines to show different tracks. The green line (Broad Gauge) is moving up, which means more of these tracks are being built. The orange line (Metre Gauge) is moving down very fast. The red line (Narrow Gauge) is very low and almost flat. This shows that the railway is switching to mostly one type of track.

Conclusion:

This chart shows that Broad Gauge is the winner in recent years. It is much higher than the other two types of tracks. We can see that the older types of tracks are slowly disappearing. Using this code and graph makes it very simple to understand how the railway is modernizing.

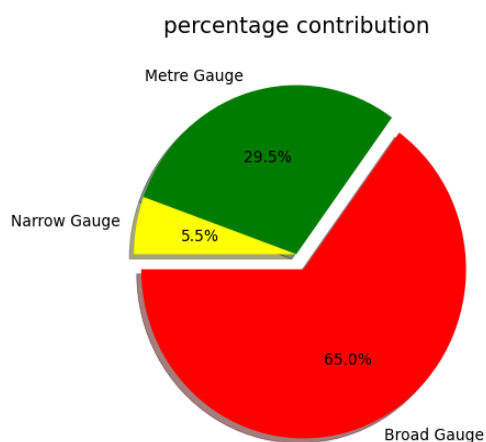
Scenario 4: Feature Engineering + Pie Chart (🟡 Medium)

Description:

In this step, we add up all the tracks from every year to see the total amount. We want to know how much each track type contributed to the whole railway history. By using a pie chart, we can see the share of each type as a "slice." This helps us understand which track type was the most common overall.

Task	Description
1.	Calculate the total sum of each gauge (Broad Gauge, Metre Gauge, Narrow Gauge) across all available years.
2.	Create a new structure (Pandas Series or single-column DataFrame) holding these three total values.
3.	Plot a pie chart using this new structure to show the percentage contribution of each gauge type.
4.	Add percentage labels to the slices using autopct (e.g., '%.1f%%').
5.	Interpret which gauge contributes the most to the entire history of the railway network.

Graph :



Graph Description:

The image shows a pie chart called "percentage contribution." It is divided into three colored slices:

- Red (Broad Gauge): This is the biggest slice at 65.0%.
- Green (Metre Gauge): This is the middle slice at 29.5%.
- Yellow (Narrow Gauge): This is the smallest slice at only 5.5%.

The red slice is pulled out slightly to show it is the most important part.

Conclusion:

This chart makes it very easy to see that Broad Gauge makes up more than half of the entire railway. By adding up all the years, we can see that it has always been the main choice. The small yellow slice shows that Narrow Gauge was a very tiny part of the system. This simple picture summarizes many years of data in one look.

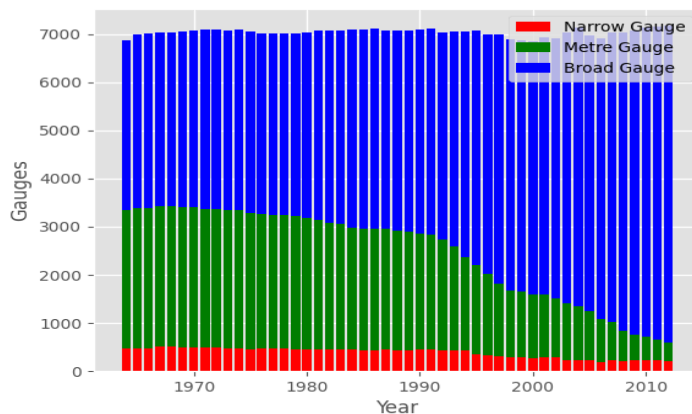
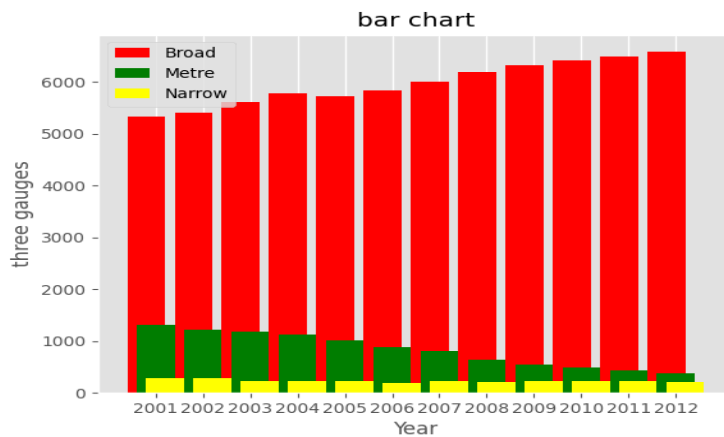
Scenario 5: Advanced Analysis + Multiple Graphs (● Hard)

Description:

This final part looks at how the railway network grows and changes every year. We calculate the percentage of each track type to see which one is winning. We also use a special tool to find the exact years when the most new tracks were built. This helps us understand the plan for the future of the railway.

Task	Description
1.	Create new columns for percentage composition relative to Total tracks for each year: % Broad Gauge, % Metre Gauge, and % Narrow Gauge.
2.	Use the NumPy function <code>np.diff</code> to calculate the yearly growth (change) of the Total tracks column.
3.	Plot: a) A single Line Graph showing the trend of all three percentage composition columns (% Broad Gauge, % Metre Gauge, % Narrow Gauge) over the years. b) A Stacked Bar Chart showing the raw composition of the three gauges (Broad, Metre, Narrow) over time.
4.	Highlight/Identify: a) The year(s) with the highest yearly growth in Total tracks. b) Any specific decade or period showing a significant decline in any particular gauge.
5.	Final Conclusion: Based on the percentage composition line graph, provide a conclusion to the question: "Is the railway system shifting towards a single dominant gauge?"

Graph :



Graph Description:

For this scenario, we use two types of charts. A line graph shows the percentages of each gauge changing over the years, making it easy to see which one is taking over. A stacked bar chart piles the different track types on top of each other. This shows both the total amount of track and how the mix of Broad, Metre, and Narrow gauges looks for every year.

Conclusion:

The data proves that the railway is moving toward using just one main track type. The Broad Gauge is growing steadily, while the Metre Gauge is quickly disappearing. By looking at the growth each year, we can see that the system is being modernized

very fast. In the end, Broad Gauge has become the single most important part of the network.

Final Conclusion:

This project shows how a big list of numbers can tell a clear story about the railway. We started by cleaning the data and then used different charts like lines, bars, and pies to see the changes. We found that over the years, the railway has moved away from older tracks to focus almost entirely on Broad Gauge. By using Python, we turned messy information into simple pictures that anyone can understand.

What you have learned :

By doing this project, I learned how to use Python to organize and fix data so it is ready to use. I practiced making different types of graphs to show information in the best way possible. I also learned how to look at a chart and explain what is happening, like seeing which parts are growing or shrinking. Overall, I now know how to use code to find real answers from a dataset.